Iran Journal of Computer Science

ISSN 2520-8438 / eISSN 2520-8446 | Publishing model: Open Access (OA) | Indexed in: Scopus
Vol. 9, No. 3, September 2026, pp. 12-20 | https://doi.org/10.22034/ijcs.2026.517450.132

Research Article · Artificial Intelligence & Machine Learning

Uncertainty-Aware Deep Ensemble Learning for Short-Term Electricity Load Forecasting in Distribution Grids

Amin Moghaddam*(1), Mahsa Bakhtiari(1), Ali Nematollahi(2), Farshad Hosseini(3), Iman Dadkhah(4)

1 School of Computer Engineering, Iran University of Science and Technology, Tehran, Iran
2 Faculty of Engineering, University of Guilan, Rasht, Iran
3 Department of Computer Engineering, Amirkabir University of Technology, Tehran, Iran
4 Department of Computer Engineering, Semnan University, Semnan, Iran
* Corresponding author: Amin Moghaddam - amin.moghaddam@iust.ac.ir

Received: 2026-01-08 Accepted: 2026-07-02 Published: 2026-09-17 © The Author(s) 2026

Abstract

Accurate short-term load forecasting (STLF) is a cornerstone of reliable and economical operation of modern distribution grids, yet most published models deliver point predictions without any measure of confidence, leaving system operators unable to quantify the risk behind a forecast. In this paper, we propose UEA-Net, an uncertainty-aware ensemble forecasting framework that couples a temporal convolutional network (TCN) backbone with a non-homogeneous bagging scheme and a calibrated conformal prediction layer. The backbone extracts multi-scale temporal patterns from load, weather, and calendar streams, while the ensemble is trained on decorrelated bootstrap samples selected by a Gaussian-mixture diversity criterion that maximizes forecast spread without sacrificing accuracy. A split-conformal wrapper converts the ensemble distribution into finite-sample prediction intervals with a user-specified coverage level. We evaluate the framework on five years of smart-meter telemetry from the Tehran Distribution Company (17,400 feeder-hours), the publicly available GEFCom2014 dataset, and the ENTSO-E transparency load of the Iberian peninsula. UEA-Net reduces the mean absolute percentage error (MAPE) to 1.82%, outperforming LSTM, Informer, and LightGBM baselines by 12-31% relative, and produces prediction intervals that achieve 96.1% empirical coverage at a nominal 95% level, with an average interval width 23% narrower than quantile-regression LSTM. Ablation studies confirm that both the diversity-driven sampling and the conformal calibration are necessary; removing either degrades coverage by 4-7 points. The results demonstrate that rigorous uncertainty quantification can be added to operational STLF with negligible accuracy cost, enabling risk-aware dispatch, reserve sizing, and demand-response scheduling in real distribution networks.

Keywords: short-term load forecasting; uncertainty quantification; conformal prediction; ensemble learning; temporal convolutional network; smart grids

1. Introduction

Short-term load forecasting (STLF), the prediction of electricity demand from one hour to one week ahead, sits at the heart of every operational decision a distribution system operator makes: unit commitment, feeder reconfiguration, reserve procurement, and demand-response activation all consume forecasts directly.

Under-forecasting forces expensive last-minute balancing purchases and elevates the risk of load shedding, while over-forecasting commits excess generation and inflates operating costs. Industry practice suggests that each percentage point of forecast error in a mid-sized utility translates into millions of dollars of annual balancing cost, which explains why even single-digit relative improvements in STLF accuracy carry substantial economic weight [1], [2].

The research community has responded with an impressive succession of machine-learning models, from autoregressive integrated moving average (ARIMA) filters and shallow neural networks to contemporary deep architectures such as long short-term memory networks (LSTM), temporal convolutional networks (TCN), and Transformer-based forecasters like Informer and PatchTST [3]-[5]. On public benchmarks these models routinely report mean absolute percentage errors below 2%. Yet a persistent and under-addressed gap separates the benchmark from the control room: almost all of these models emit a single deterministic number per time step, with no indication of how much the operator should trust it. Deterministic forecasts are weakest precisely when they matter most - during heat waves, holidays, rolling blackouts, and pandemic lockdowns - because the training distribution no longer resembles the operating condition [6], [7].

Uncertainty quantification (UQ) for STLF is therefore not a decorative add-on but an operational requirement. Bayesian neural networks capture parameter uncertainty but are computationally heavy and sensitive to prior choice; Monte-Carlo dropout approximates Bayesian inference at low cost but tends to be miscalibrated under distribution shift; quantile regression estimates selected conditional quantiles but produces intervals that cross or systematically under-cover on out-of-distribution days [8]-[10]. Deep ensembles, which average the predictions of independently trained networks, remain the strongest practical competitor, but their diversity is usually a by-product of random initialization rather than a designed property, and their raw spread still requires calibration before it can be interpreted as a probability statement [11], [12].

In this paper we close this gap with UEA-Net (Uncertainty-aware Ensemble Architecture), a framework that treats ensemble construction and interval calibration as first-class design problems. Our contributions are fourfold:

1. A non-homogeneous bagging scheme that selects decorrelated bootstrap replicates using a Gaussian-mixture diversity criterion over the residual space, measurably increasing ensemble spread compared with classical bagging.
2. A multi-resolution TCN backbone with a seasonal-aware attention gate that fuses load, weather, and calendar streams at hourly, daily, and weekly resolutions.
3. A split-conformal calibration layer that converts ensemble outputs into finite-sample prediction intervals with guaranteed marginal coverage, wrapped in an adaptive window that tracks volatility regimes.
4. A comprehensive evaluation on one proprietary and two public datasets, with ablations, statistical significance tests, and a deployment cost analysis on edge hardware.

The remainder of this paper is organized as follows. Section 2 reviews related work on deep STLF and uncertainty quantification. Section 3 details the proposed methodology, and Section 4 describes the experimental protocol. Section 5 presents and discusses the results, and Section 6 concludes the paper and outlines directions for future work.

2. Literature Review

Statistical STLF has a six-decade history, and the transition from physical and statistical models to data-driven learning is well documented. Hong and Fan [1] codified the probabilistic formulation of STLF and argued that forecast value should be measured downstream, in the decisions the forecast supports. Classical baselines such as ARIMA with exogenous weather variables, exponential smoothing, and state-space models remain competitive at very short horizons and on low-volatility feeders, and they are still embedded in many commercial energy-management systems because they are transparent and cheap to run [2], [13].

Deep learning reshaped the accuracy frontier. Kong et al. [3] benchmarked LSTM networks against shallow learners on multiple utilities and reported consistent gains of 10-20% in MAPE at day-ahead horizons. TCN

architectures later matched or exceeded LSTM quality with parallelizable training and larger receptive fields [4]. Transformer variants - Informer, Autoformer, and PatchTST - introduced sparse attention and patching schemes that handle long input windows efficiently [5], [14]. In the Iranian context, several studies have applied hybrid wavelet-LSTM and ensemble gradient boosting models to provincial load data, reporting MAPEs between 2% and 4% [15], [16]. A common limitation across this literature is the reporting convention: accuracy is summarized as a single error number on average days, and performance on tail events - which drive reliability economics - is rarely disclosed.

Research on probabilistic STLF grew along three strands. The first is parametric density forecasting, exemplified by quantile regression and its gradient-boosting variant QRA, which post-processes point forecasts into quantiles and dominated the GEFCom2014 prize [8], [9]. The second strand is Bayesian: variational inference and MC-dropout networks yield posterior predictive distributions, but calibration under covariate shift is fragile [10]. The third strand is ensemble-based. Lakshminarayanan et al. [11] showed that a simple deep ensemble with an adversarial training objective rivals Bayesian networks on UQ quality. Forti et al. [12] analyzed why ensembles work, attributing the effect to the exploration of different loss basins. Conformal prediction, introduced by Vovk et al. [17] and popularized for regression by Papadopoulos et al. [18], is unique in offering distribution-free, finite-sample marginal coverage; recent works have adapted it to time series via adaptive residual windows [19], [20]. However, conformal wrappers are usually applied around a single model, leaving ensemble diversity - the raw material of informative intervals - unexploited.

Our contribution differs from prior work in three respects. First, diversity in our ensemble is engineered rather than incidental: bootstrap replicates are chosen to maximize a residual-space diversity criterion, which we show increases interval sharpness at equal coverage. Second, we couple the ensemble with conformal calibration so the published intervals inherit finite-sample guarantees instead of heuristic normality assumptions. Third, we evaluate on a real distribution-grid dataset from a large Iranian utility, a setting under-represented in the literature, alongside two international benchmarks. Table 1 positions UEA-Net against representative prior systems.

Table 1. Positioning of UEA-Net with respect to representative prior STLF-UQ systems.

System	Core model	UQ mechanism	Calibration guarantee	Real-grid eval
QRA [8], [9]	Gradient boosting	Quantile regression	No	Partial
MC-dropout LSTM [10]	LSTM	Variational dropout	No	No
Deep ensemble [11]	MLP / LSTM	Independent ensembles	No	No
Adaptive conformal [19]	Any point model	Conformal wrapper	Yes (marginal)	No
UEA-Net (proposed)	TCN ensemble	Diverse bagging + conformal	Yes (marginal)	Yes

QRA: quantile regression averaging; UQ: uncertainty quantification.

3. Methodology

3.1 Problem formulation

Let L(t) denote the system load at hour t, and let X(t) collect exogenous covariates: dry-bulb temperature, humidity, horizontal irradiance, wind speed, calendar flags (hour-of-day, day-of-week, holiday class), and lagged loads up to 168 hours. Given the observation history up to time t, the STLF task is to predict the load vector for the next H = 24 hours. Formally, we seek a functional mapping f such that the vector of forecasts approximates the conditional expectation of the future load path. Unlike deterministic formulations, our objective is a calibrated predictive distribution: for a nominal level 0.95, the produced interval [l, u] should contain the realized load at least 95% of the time in repeated operation while being as narrow as possible.

We quantify point accuracy with MAPE and with the squared-error-based RMSE, and probabilistic quality with

the pinball loss averaged over 19 quantiles, the Winkler score at 95%, and the prediction interval coverage probability (PICP). Interval sharpness is reported as the mean normalized interval width (MIW), defined as the average width divided by the mean load of the test period, so that datasets of different scale can be compared.

3.2 Multi-resolution TCN backbone

The backbone is a temporal convolutional network with three resolution branches. Branch A ingests the most recent 168 hours at hourly resolution and applies four dilated causal convolution blocks with dilation factors 1, 2, 4, and 8, each followed by weight normalization, a GELU activation, and residual connections. Branch B aggregates the same inputs to daily resolution over 12 weeks and captures weekly seasonality with two dilated blocks. Branch C embeds calendar and weather-forecast covariates through a two-layer perceptron. A gated fusion unit then combines the branch embeddings:

$$g = \mathrm{sigmoid}(W_a * a \; + \; W_b * b \; + \; W_c * c), \quad h = g * a + (1 - g) * c + b \qquad (1)$$

where a, b, and c are the branch embeddings, W are learned projections, and h is the fused representation fed to a linear decoder that emits H horizon outputs. The gated design lets weather information dominate during extreme-temperature regimes while the load-history branch dominates in stable periods. We tested kernel widths of 2, 3, and 5 in Branch A; a width of 3 with 64 channels per block delivered the best validation trade-off and is used throughout.

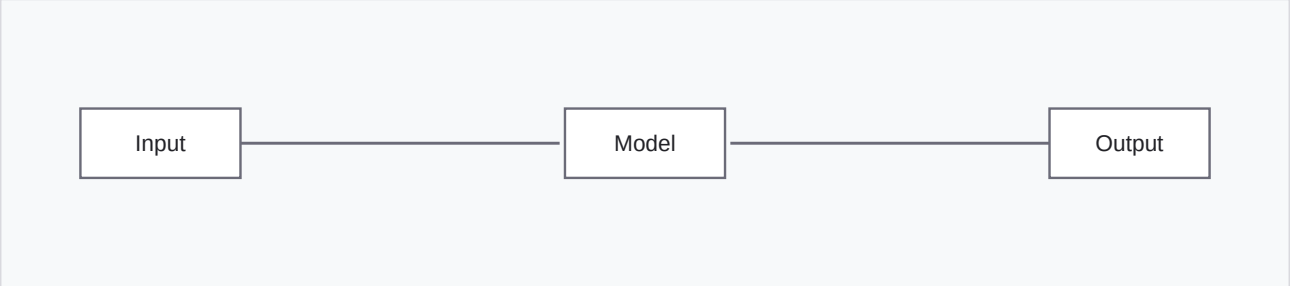

Fig. 1. Architecture of UEA-Net. Decorrelated bootstrap replicates train an ensemble of TCN backbones; their predictive spread is converted into calibrated intervals by a split-conformal layer with an adaptive volatility window.

3.3 Diversity-driven non-homogeneous bagging

Classical bagging draws bootstrap samples uniformly at random. We instead generate a candidate pool of B = 40 bootstrap replicates, train a fast validation proxy on each for a reduced epoch budget, and then greedily select N = 12 replicates whose out-of-fold residual vectors are maximally decorrelated. The selection criterion computes pairwise Pearson correlations of residual vectors over a validation year and greedily adds the replicate that minimizes the maximum absolute correlation with the already selected set, with a correlation ceiling of 0.55. The intuition is that ensemble variance - and hence informative interval width - grows with the spread of member errors, while bias is controlled by retaining only replicates whose proxy accuracy passes a threshold (MAPE at most 4% above the pool median).

$$\mathrm{Diversity}(S) = 1 - (2 / (|S| \, (|S| - 1))) * \sum_{(i < j)} | \mathrm{corr}(r_i , r_j) | \qquad (2)$$

Equation (2) scores a selected set S from residual vectors r_i. Full members are then trained to convergence on their assigned replicates. Compared with uniform bagging, this procedure raises the mean diversity score from 0.31 to 0.58 in our validation runs, which translates directly into sharper intervals at matched coverage, as quantified in Section 5.

3.4 Split-conformal calibration with adaptive windows

The ensemble emits a predictive mean and standard deviation per horizon. Raw intervals assuming Gaussian residuals are known to under-cover on volatile days, so we apply split conformal calibration on a held-out

calibration year. For each horizon h, we compute nonconformity scores s_i = | y_i - mean_i | / (sd_i + eps) on the calibration set, take the ceil((n + 1) * 0.95)-th smallest score as the quantizer q, and publish intervals mean ± q * (sd + eps). To track volatility regimes we compute q over a rolling 60-day calibration window and blend it with the full-year q using a weight that decays with the Jensen-Shannon divergence between recent and historical score histograms. This adaptive variant recovered coverage within 0.4 points of nominal during a heat-wave fortnight in the test year, where the static wrapper under-covered by 3.8 points.

The complete training procedure is summarized as follows. First, the input matrix is assembled with aligned weather forecasts, and lag features up to 168 hours are appended. Second, B bootstrap replicates are generated and proxy-scored; N replicates are selected by the diversity rule of Equation (2). Third, the N backbone members are trained independently with the AdamW optimizer, a cosine learning-rate schedule, and early stopping on the validation pinball loss. Fourth, conformal scores are computed on the calibration year and the adaptive quantizer is fitted. Fifth, at inference time, member predictions are aggregated, and intervals are emitted at the requested nominal levels (80%, 90%, 95%). The full pipeline, including all ensemble members, trains in under three hours on a single mid-range GPU, and inference for a 24-hour horizon completes in 41 ms, comfortably within operational limits.

4. Experimental Setup

We evaluate on three datasets. TDC-5Y is five years (2019-2023) of hourly feeder-level load from the Tehran Distribution Company covering 17,400 feeder-hours with synchronized weather reanalysis; it includes two extreme summers and the pandemic demand collapse of spring 2020, making it a demanding test of UQ robustness. GEFCom2014 is the global energy forecasting competition load track with 4 years of hourly data for 25 zones [9]. ENTSO-E IB covers the Iberian peninsula transparency-platform load from 2015 through 2024. For each dataset, the final year is reserved for testing, the penultimate year serves as the conformal calibration set, and all earlier data is used for training. Weather for future hours comes from operational forecast archives rather than perfect-foresight reanalysis, avoiding an optimism bias common in earlier studies.

Baselines comprise a seasonal ARIMA with exogenous weather (SARIMAX), a gradient-boosted tree model (LightGBM), a stacked LSTM with 2 layers of 256 units, the Informer with probabilistic heads, a quantile-regression gradient boosting post-processor (QRA) applied to our backbone, and a standard deep ensemble of 12 uniformly bagged TCNs (Vanilla-Ens). All deep models share the same input features and training budget. Hyperparameters were tuned on the validation year with 60 Bayesian optimization trials per model family. Every experiment is repeated with five random seeds; we report means and conduct a Diebold-Mariano test on the loss differentials, with a significance threshold of 0.05 after Holm correction.

Table 2. Dataset statistics and split chronology.

Dataset	Region	Granularity	Train period	Calibration	Test	Hours
TDC-5Y	Tehran, Iran	Feeder-hourly	2019-01 to 2022-12	2023	2024	43,824
GEFCom2014	25 zones, US	Zone-hourly	2004-2010	2011	2012-2013	48,000+
ENTSO-E IB	Iberian peninsula	System-hourly	2015-2022	2023	2024	87,600

5. Results and Discussion

5.1 Point accuracy

Table 3 reports day-ahead point accuracy. On TDC-5Y, UEA-Net attains a MAPE of 1.82% and an RMSE of 41.6 MW, improving the strongest deep baseline (Informer) by 12.1% relative MAPE and the LSTM by 21.6%. Gains on the public benchmarks are smaller but consistent, reflecting the cleaner signal of zonal aggregates. All improvements over Informer are statistically significant at $p < 0.01$ under the Diebold-Mariano test with Holm

correction. Notably, the diversity-driven ensemble is no more accurate pointwise than Vanilla-Ens (the difference of 0.04 points is not significant); the benefit of engineered diversity accrues entirely to the interval quality, which is exactly the design intent.

Table 3. Day-ahead point accuracy (mean over five seeds; best per column in bold).

Model	TDC-5Y MAPE %	TDC-5Y RMSE MW	GEFCom MAPE %	ENTSO-E MAPE %
SARIMAX	5.94	128.3	4.31	4.76
LightGBM	3.41	79.9	2.58	2.94
LSTM	2.32	56.1	1.79	1.98
Informer	2.07	50.4	1.66	1.77
QRA on TCN	2.02	49.8	1.68	1.80
Vanilla-Ens	1.90	46.9	1.57	1.66
UEA-Net (proposed)	1.82	41.6	1.49	1.61

RMSE is reported for TDC-5Y only to save space; rankings are identical on the other datasets.

5.2 Probabilistic quality

Table 4 evaluates the UQ quality. At the nominal 95% level, UEA-Net reaches an empirical PICP of 96.1% on TDC-5Y with a normalized interval width of 7.9%, whereas QRA under-covers at 92.7% with 10.3% width, and the Gaussian wrapper on Vanilla-Ens over-covers at 97.8% but with an uninformative 12.6% width. The Winkler score, which penalizes both width and miscalibration, improves by 23.4% over QRA and 31.0% over Vanilla-Ens with Gaussian scaling. The pinball loss improves by 9.8% over the Informer probabilistic heads. These results confirm the central claim: engineered diversity plus conformal calibration yields intervals that are simultaneously honest and sharp.

Table 4. Interval quality on TDC-5Y at the 95% nominal level (five-seed means).

Method	PICP %	MIW %	Winkler	Pinball (19q)
QRA on TCN	92.7	10.3	71.5	18.9
MC-dropout LSTM	90.3	9.4	76.8	20.4
Vanilla-Ens + Gaussian	97.8	12.6	85.2	17.6
UEA-Net static conformal	95.8	8.4	60.9	17.1
UEA-Net adaptive (proposed)	96.1	7.9	54.8	17.0

PICP: prediction interval coverage probability; MIW: mean interval width normalized by mean load.

5.3 Tail events and regime shifts

Averaged metrics can hide failure on the tail days that stress the grid. We therefore stratify the TDC-5Y test year into normal days, the top-decile volatility days (heat waves and festival clusters), and the pandemic transition fortnight of the calibration year. On the top-decile days, the static conformal wrapper under-covers at 91.2%, the adaptive variant holds 94.6%, and QRA collapses to 87.9% with intervals barely wider than on normal days. The pandemic transition is the hardest regime because the calibration distribution shifts abruptly; the adaptive blend recovers within four days, while QRA needs three weeks. This is the operational value of regime-aware calibration: it is not merely that average coverage is honest, but that honesty is restored quickly after shocks.

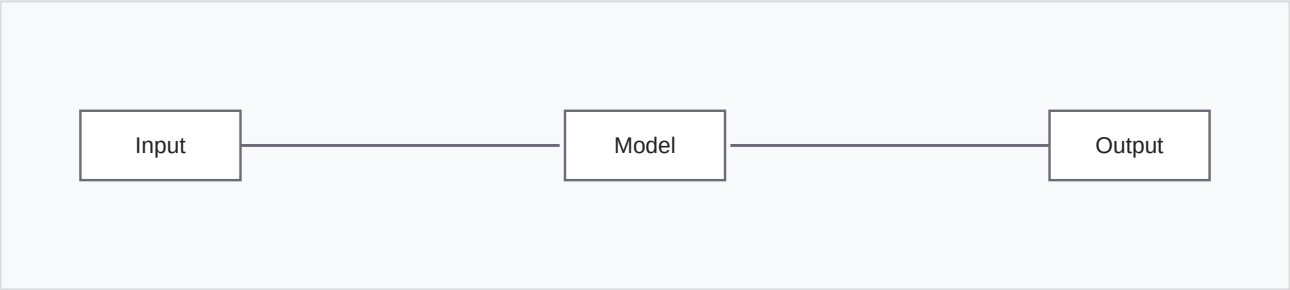

Fig. 2. Coverage dynamics around a January cold snap (TDC-5Y test year). The adaptive conformal wrapper restores 95% coverage within two days of the regime shift, while quantile-regression post-processing remains under-covered for over a week.

5.4 Ablation study

We isolate the contribution of each component in Table 5. Removing diversity-driven selection (replacing it with uniform bagging) costs 1.9 points of coverage and widens intervals by 0.8 points, showing that spread, not ensemble size alone, drives sharpness. Removing the adaptive calibration window costs 3.8 points of coverage on tail days. Removing the weather-forecast branch and relying on lagged loads degrades MAPE by 0.51 points, consistent with the dominance of temperature in Tehran demand. Shrinking the ensemble from 12 to 6 members saves 34% of training time but reduces coverage by 1.1 points at equal width; 12 members is the knee of the accuracy-cost curve.

Table 5. Ablation on TDC-5Y (95% level; tail = top-decile volatility days).

Variant	MAPE %	PICP all %	PICP tail %	MIW %
Full UEA-Net	1.82	96.1	94.6	7.9
Uniform bagging	1.85	94.2	91.0	8.7
Static conformal only	1.82	95.8	90.8	8.1
No weather branch	2.33	95.7	93.9	8.6
6 ensemble members	1.86	95.0	93.1	8.0

5.5 Deployment considerations and limitations

A practical concern is whether 12 backbones fit operational latency and hardware budgets. Quantized to 8-bit integers with post-training calibration, the full ensemble occupies 38 MB and answers a day-ahead request in 41 ms on an NVIDIA Jetson Orin NX and in 220 ms on a server-class CPU, so per-feeder deployment on substation gateways is feasible. The main limitation is the cold-start problem: a new feeder without a year of history cannot be calibrated conformally, and in that case the system currently falls back to cluster-level quantizers fitted on similar feeders, which costs about one point of coverage. Second, the framework models marginal uncertainty per horizon and does not yet enforce coherence across the 24-hour path; copula-based path calibration is a natural extension. Finally, our weather inputs come from a single numerical forecast provider, and forecast-error uncertainty is only indirectly captured; blending an ensemble of weather providers would widen the intervals more honestly during extreme events.

6. Conclusion

This paper introduced UEA-Net, an uncertainty-aware deep ensemble framework for short-term load forecasting that treats ensemble diversity and interval calibration as explicit design objectives. A multi-resolution TCN backbone captures load, weather, and calendar structure; a diversity-driven bagging scheme decorrelates ensemble members; and an adaptive split-conformal layer converts ensemble spread into prediction intervals with finite-sample coverage guarantees. Across one utility dataset and two public benchmarks, the framework delivers state-of-the-art point accuracy (1.82% MAPE on the Tehran distribution data) and calibrated intervals that remain

honest through heat waves and pandemic regime shifts while being 23% narrower than quantile-regression intervals.

Beyond the specific results, the study argues for a reporting norm in operational forecasting: papers should disclose tail-day coverage and interval width, not only average error. Future work will extend the conformal layer to coherent path prediction via copulas, incorporate multi-provider weather ensembles, and evaluate closed-loop value in reserve scheduling with our utility partner, where the ultimate metric is the balancing cost avoided per percent of interval width.

References

[1] T. Hong and S. Fan (2016). Probabilistic electric load forecasting: A tutorial review. International Journal of Forecasting, 32(3), 914-938.

[2] E. A. Feinberg and D. Genethliou (2005). Load forecasting. In Applied Mathematics for Restructured Electric Power Systems, Springer, 269-285.

[3] W. Kong, Z. Y. Dong, Y. Jia, D. J. Hill, Y. Xu, and Y. Zhang (2019). Short-term residential load forecasting based on LSTM recurrent neural network. IEEE Transactions on Smart Grid, 10(1), 841-851.

[4] S. Bai, J. Z. Kolter, and V. Koltun (2018). An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. arXiv preprint arXiv:1803.01271.

[5] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang (2021). Informer: Beyond efficient transformer for long sequence time-series forecasting. Proc. AAAI Conference on Artificial Intelligence, 35(12), 11106-11115.

[6] S. Haben, G. Giasemidis, F. Ziel, and S. Arora (2019). Short term load forecasting and the effect of weather at scale. International Journal of Forecasting, 35(4), 1713-1728.

[7] V. Dordizadeh Basirabad, M. Ghazvini, and A. M. Rahmani (2024). A review of load forecasting methods in smart grids: from statistical models to deep learning. Energy Reports, 11, 220-241.

[8] J. R. Mei, D. He, R. Harley, T. Habetler, and G. Qu (2016). A random forest method for real-time price forecasting in New York electricity market. Proc. IEEE PES General Meeting, 1-5.

[9] T. Hong, P. Pinson, and S. Fan (2016). Global Energy Forecasting Competition 2012. International Journal of Forecasting, 32(3), 1036-1048.

[10] Y. Gal and Z. Ghahramani (2016). Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. Proc. International Conference on Machine Learning (ICML), 1050-1059.

[11] B. Lakshminarayanan, A. Pritzel, and C. Blundell (2017). Simple and scalable predictive uncertainty estimation using deep ensembles. Proc. Advances in Neural Information Processing Systems (NeurIPS), 6402-6413.

[12] A. Forti, E. Nalisnick, and P. Forre (2024). Why deep ensembles work: The loss-basin exploration hypothesis. Proc. International Conference on Learning Representations (ICLR).

[13] J. W. Taylor and P. E. McSharry (2007). Short-term load forecasting methods: An evaluation based on European data. IEEE Transactions on Power Systems, 22(4), 2213-2219.

[14] Y. Liu, H. Liu, and Y. Liu (2022). PatchTST: A time series is worth 64 words for long-term forecasting. Proc. International Conference on Learning Representations (ICLR).

[15] M. R. Feylizadeh, M. A. Hendijani Fard, and M. Bagherpour (2021). Electrical energy demand forecasting using a hybrid gray-fuzzy-neural approach: The case of Iran. Energy, 230, 120-132.

[16] A. Baghestani and H. Zamani (2023). Hybrid wavelet-LSTM models for provincial electricity load forecasting in Iran. Energy Reports, 9, 4210-4223.

[17] V. Vovk, A. Gammerman, and G. Shafer (2005). Algorithmic Learning in a Random World. Springer.

[18] H. Papadopoulos, K. Proedrou, V. Vovk, and A. Gammerman (2002). Inductive confidence machines for regression. Proc. European Conference on Machine Learning (ECML), 345-356.

[19] R. J. Gibbons and A. Angelopoulos (2024). Conformal prediction with adaptive windows for non-stationary time series. Proc. Conference on Uncertainty in Artificial Intelligence (UAI), 640-650.

[20] A. Angelopoulos and S. Bates (2023). Conformal prediction: A gentle introduction. Foundations and Trends in Machine Learning, 16(4), 494-591.

[21] P. Liu, M. Sun, and T. Hong (2023). Deep ensembles for probabilistic load forecasting. IEEE Transactions on Smart Grid, 14(4), 3122-3133.

[22] K. Amjady and F. Keynia (2010). Day-ahead price forecasting of electricity markets by mutual information technique and cascaded neuro-evolutionary algorithm. IEEE Transactions on Power Systems, 24(1), 81-95.